Distinguishing Goppa codes using higher-order vanishing

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Abstract. We present a new algorithm to distinguish alternant and Goppa codes from general linear codes. Our approach is based on higher-order vanishing of polynomials and can be applied to a wide set of code parameters. It also applies to Goppa code parameters used in the Classic McEliece key encapsulation mechanism. While these parameters are out of reach in practice, we analyse the behaviour of our distinguisher and estimate its complexity for them, indicating that it is more efficient than previous distinguishing approaches. This is supported by concrete experiments that distinguish codes with larger parameters in practice.

1 Introduction

The McEliece encryption scheme [19] is one of the oldest public-key cryptosystems. Since its publication in 1978, its security has remained remarkably stable [2]. Unlike the RSA cryptosystem, which was published a few years earlier, it is deemed secure not only against classical algorithms but also against large-scale quantum computers. The Classic McEliece key encapsulation mechanism [1] was a fourth-round candidate in the NIST standardisation process for post-quantum cryptography. It was not selected for standardisation by NIST mostly due to its large public key size, which limits its usability in many applications [3].

The McEliece cryptosystem employs binary Goppa codes, a family of error-correcting codes for which efficient decoding algorithms are known. Roughly speaking, the security of the (Classic) McEliece scheme is based on the assumption that decoding random errors for a randomly selected binary Goppa code is hard if one is given only a random basis of the Goppa code. The best known algorithms to solve this problem are information-set decoding algorithms [22][25][18][6], whose complexity has remained very stable over the past 60 years.

One way to approach the study of the hardness assumption that underpins the security of the McEliece cryptosystem is to consider a related but potentially easier problem. It is known that the decoding problem for general linear codes is NP-complete [8] and it is also believed to be hard on average. A natural question to ask is: Given a random basis (a) of a randomly selected (binary) Goppa code, or (b) of a randomly selected linear code of a fixed dimension, can one determine if one has been given (a) or (b)? We will call this the *distinguishing problem for Goppa codes*. If this problem is hard, then the problem underlying the security of (Classic) McEliece is also hard. The converse need not be true, of course.

Previous work. Due to its relationship with the McEliece cryptosystem, the distinguishing problem has received some attention over the past 15 years. [12] first describes a polynomial-time distinguisher for alternant/Goppa codes that works if the codes have a sufficiently high rate. The distinguisher exploits the fact that a certain linear system obtained by linearizing a polynomial system describing key recovery behaves differently when it is constructed from a random code or an alternant/Goppa code. [21] observes that this distinguisher has a simpler interpretation in terms of componentwise products of codes. We refer to this as the *square distinguisher* and will briefly sketch its main idea below. [20] proves tight bounds for the behaviour of this distinguisher in the Goppa and alternant case. Hence, the authors are able to determine precisely the parameter ranges in which Goppa or alternant codes can be efficiently distinguished from random codes with high probability using the square distinguisher. A new approach using quadratic forms [10] allows in principle to distinguish alternant/Goppa codes outside the high-rate regime including in parameter ranges used in the McEliece cryptosystem. Its complexity is estimated to be lower than the best known key recovery attack against the McEliece cryptosystem, but it still grows rapidly outside the high-rate regime. Finally, the syzygy distinguisher [23] uses higher-order Betti numbers as invariants to distinguish alternant/Goppa and random codes. It is the first distinguisher that is asymptotically sub-exponential in the error-correcting capability of the code.

Outline of our approach. We introduce a new approach to distinguish alternant/Goppa codes and random codes that is based on higher-order vanishing of polynomials. Similarly to previous distinguishers, our starting point is the square distinguisher, which we briefly recall. Given a linear code $C \subset \mathbb{F}_q^n$, we define the *square code* of C as

$$C^{*2} \coloneqq \langle c * d \mid c, d \in C \rangle$$

where $c*d \coloneqq (c_1 d_1, \ldots, c_n d_n)$ denotes the componentwise product of two vectors and $\langle \cdot \rangle$ denotes the $\mathbb{F}_q$-span. If C is a linear code of dimension k, then clearly

$$\dim_{\mathbb{F}_q} \left(C^{*2} \right) \leq \min \left\{ n, \binom{k+1}{2} \right\}$$

If C is randomly chosen of sufficiently large dimension k, then [9] shows that the dimension of its square code attains this maximal value with high probability. In contrast, if $C = \mathrm{GRS}_k(\boldsymbol{x}, \boldsymbol{y})$ is a generalised Reed-Solomon code (see Definition 1), it is easy to see that

$$C^{*2} = \mathrm{GRS}_{2k-1}(\boldsymbol{x}, \boldsymbol{y} * \boldsymbol{y})$$

and so its dimension is at most $2k-1$. If $2k-1 < n$, we can therefore detect a difference with high probability and thus distinguish GRS codes from random codes. It is known that the dual of a GRS code is again a GRS code. So, if $2k - 1 \geq n$, we can instead compute the dimension of the square of the *dual* of the given code. Thus, the square distinguisher allows to distinguish GRS codes from random codes of the same dimension with high probability.

Suppose now that $C = \mathrm{Alt}_r(\boldsymbol{x}, \boldsymbol{y})$ is an alternant (or, more specifically, a Goppa) code over $\mathbb{F}_q$ with field extension degree m. The authors of [20] prove upper bounds on the dimension of $\left(C^\perp\right)^{*2}$ in this case. Note that by Proposition 1 below, after field extension (which does not affect the dimension of square codes), $C^\perp$ contains GRS codes. This is the reason that the dimension of the square code is expected to be smaller than for random linear codes. However, for $m > 1$ the square distinguisher works only for Goppa (or alternant) codes of high rate (so that its dual has low rate). This is because if we lower the rate of C, then $\left(C^\perp\right)^{*2} = \mathbb{F}_q^n$ soon fills the whole space in both the alternant/Goppa and the random case, so they cannot be distinguished.

The square distinguisher may equivalently be described as follows: Let $\boldsymbol{A} \in \mathbb{F}_q^{k\times n}$ be a matrix. We are interested in the case that $\boldsymbol{A}$ is the parity check matrix of an alternant/Goppa or random linear code. Let $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n \in \mathbb{F}_q^k$ denote the columns of $\boldsymbol{A}$. Denoting by $\mathbb{F}_q[X_1, \dots, X_k]^{(p)}$ the homogeneous polynomials of degree p, we define the map

$$\varphi_{\boldsymbol{A}}^{(2)} \colon \mathbb{F}_q[X_1, \dots, X_k]^{(2)} \to \mathbb{F}_q^n, \quad f \mapsto (f(\boldsymbol{a}_1), \dots, f(\boldsymbol{a}_n))$$

Note that the image of $\varphi_{\boldsymbol{A}}^{(2)}$ is precisely the square of the code with generator matrix $\boldsymbol{A}$. Thus, the square distinguisher essentially considers the rank (or, equivalently, the dimension of the kernel) of the map $\varphi_{\boldsymbol{A}}^{(2)}$.

A natural generalisation of this map is to not only consider quadratic polynomials in the domain, but also polynomials of higher degree. This is equivalent to considering C^{*m} for $m \geq 3$. However, for our problem of distinguishing Goppa/alternant codes from random codes, this does not yield additional information compared to the square distinguisher: If the code C^{*3} is a proper subset of $\mathbb{F}_q^n$ (which is required to be able to distinguish C from a random code), then so is C^{*2}. Thus the square distinguisher is already sufficient.

Instead of only considering simple vanishing of homogeneous polynomials at columns of $\boldsymbol{A}$, our main idea is to consider homogeneous polynomials of higher degree whose vanishing at the columns of $\boldsymbol{A}$ is *of higher order*. For example, we compute homogeneous polynomials of degree 3 that do not only themselves vanish at all columns of $\boldsymbol{A}$, but so do their first-order partial derivatives. More formally, our distinguisher (in degree 3) computes the dimension of the kernel of the map

$$\varphi_{\boldsymbol{A}}^{(3)} \colon \mathbb{F}_q[X_1, \dots, X_k]^{(3)} \to \mathbb{F}_q^{n(k+1)}, \quad f \mapsto \left[f(\boldsymbol{a}_j), \frac{\partial f}{\partial X_i}(\boldsymbol{a}_j)\right]_{i=1,\dots,k}^{j=1,\dots,n}$$

in the cases that $\boldsymbol{A}$ is the parity check matrix of an alternant/Goppa or random linear code. This can be extended to higher degrees: In degree p, our distinguisher computes the dimension of the space of homogeneous polynomials of degree p whose partial derivatives of order up to $p-2$ all vanish when evaluated at the columns of $\boldsymbol{A}$.

Summary of results. We propose a new approach for distinguishing Goppa (or, more generally, alternant) codes from random linear codes using higher-order vanishing. We analyse our distinguisher and explain its behaviour in the Goppa (or alternant) and random case. Our experiments demonstrate the effectiveness of the distinguisher for concrete parameter sets. To our knowledge, this approach is capable of distinguishing larger parameter sets in practice than previous approaches. Based on our analysis, we predict the complexity of distinguishing binary Goppa codes with parameters proposed in the Classic McEliece key encapsulation mechanism using our distinguisher. Although still too large to distinguish these Goppa codes in practice, the complexity is lower than previous approaches.

Organisation of this paper. In section 2, we list some basic definitions and results in coding theory as well as notation for differentiation of polynomials. In section 3, we introduce a map that encodes simultaneous higher-order vanishing of polynomials at a given set of points and prove some basic properties. This is the central object of study in this paper. In section 4 and section 5, we analyse its behaviour for random and alternant/Goppa codes, respectively. These results allow us to specify our distinguisher and predict its behaviour in section 6. In subsection 6.1, we provide experimental results for concrete parameter sets. In subsection 6.2, we estimate the complexity for Classic McEliece parameter sets. We close with some remarks on similarities and differences compared to the syzygy distinguisher from [23] in subsection 6.3.

Notation and conventions. Throughout this paper, let q be a prime power. Given an integer k, we denote by $R_k := \mathbb{F}_q[X_1, \ldots, X_k]$ the polynomial ring in k indeterminates over $\mathbb{F}_q$. For any non-negative integer p, we denote by $R_k^{(p)} \subset R_k$ the vector space of homogeneous polynomials of degree p. We denote matrices by boldface upper-case letters and vectors as boldface lower-case letters.

2 Preliminaries

2.1 Linear codes

Recall that a linear $[n, k]$-code over $\mathbb{F}_q$ is a k-dimensional subspace $C \subset \mathbb{F}_q^n$. In this work, we are mainly interested in alternant and Goppa codes, which are closely related to generalised Reed-Solomon codes. For convenience, we recall their definitions.

Definition 1. *Suppose $x_i, y_i \in \mathbb{F}_q$ for $1 \leq i \leq n$ with x_i pairwise distinct and $y_i \neq 0$ for all i. The* generalised Reed-Solomon (GRS) code *of dimension r is defined by*

$$GRS_r(\boldsymbol{x}, \boldsymbol{y}) := \{(y_1 f(x_1), \ldots, y_n f(x_n)) \mid f \in \mathbb{F}_q[X],\ deg(f) < r\}$$

Alternant codes are defined by taking a GRS code over a larger field $\mathbb{F}_{q^m}$ and restricting to those vectors whose entries all lie in the subfield $\mathbb{F}_q$. Thus, an alternant code is a subfield subcode of a GRS code:

Definition 2. *Let $\boldsymbol{x} \in \mathbb{F}_{q^m}^n$ have distinct entries and $\boldsymbol{y} \in (\mathbb{F}_{q^m}^*)^n$. The* alternant code *with* support $\boldsymbol{x}$ *and* multiplier $\boldsymbol{y}$ *and* degree r *over* $\mathbb{F}_q$ *is*

$$Alt_r(\boldsymbol{x}, \boldsymbol{y}) \coloneqq GRS_r(\boldsymbol{x}, \boldsymbol{y})^{\perp} \cap \mathbb{F}_q^n$$

Such an alternant code is proper *if it has dimension $n - mr$.*

Note that in the definition, we define alternant codes to be the subfield subcode of the *dual* of a GRS code. However, this is mere convention since duals of GRS codes are again GRS codes with the same support but a different multiplier [16, Ch. 10, Theorem 4].

Goppa codes are alternant codes with a special choice of multiplier:

Definition 3. *Let $\boldsymbol{x} \in \mathbb{F}_{q^m}^n$ have distinct entries and $\gamma \in \mathbb{F}_{q^m}[X]$ be a polynomial of degree t with $\gamma(x_i) \neq 0$ for all $1 \leq i \leq n$. The* Goppa code *with* support $\boldsymbol{x}$ *and* Goppa polynomial γ *is defined as*

$$Gop(\boldsymbol{x}, \gamma) \coloneqq Alt_t\left(\boldsymbol{x}, \boldsymbol{y}\right)$$

where $\boldsymbol{y} \coloneqq \left(\frac{1}{\gamma(x_1)}, \ldots, \frac{1}{\gamma(x_n)}\right)$.

For the purposes of our distinguisher, we only need the following property of alternant codes which is proved in [5, Proposition 14].

Proposition 1. *Let $\boldsymbol{x} \in \mathbb{F}_{q^m}^n$ have distinct entries, $\boldsymbol{y} \in (\mathbb{F}_{q^m}^*)^n$, and $Alt_r(\boldsymbol{x}, \boldsymbol{y})$ be an alternant code over $\mathbb{F}_q$. Then*

$$Alt_r(\boldsymbol{x}, \boldsymbol{y})^{\perp} \otimes \mathbb{F}_{q^m} = \sum_{i=0}^{m-1} GRS_r\left(\boldsymbol{x}^{q^i}, \boldsymbol{y}^{q^i}\right)$$

where $\boldsymbol{c}^{q^i} \coloneqq (c_1^{q^i}, \ldots, c_n^{q^i})$ denotes component-wise powers for any $\boldsymbol{c} \in \mathbb{F}_{q^m}^n$.

2.2 Differentials

Recall that we denote by $R_k = \mathbb{F}_q[X_1, \ldots, X_k]$ the polynomial ring over $\mathbb{F}_q$ with k indeterminates. Given $g \in R_k$, we define the differential operator $\mathcal{D}_g$ as

$$\mathcal{D}_g \colon R_k \to R_k, \quad f \mapsto g\left(\frac{\partial}{\partial X_1}, \ldots, \frac{\partial}{\partial X_k}\right) \cdot f$$

This is a natural extension of (higher-order) partial derivatives. To illustrate the definition with examples:

- $\mathcal{D}_{X_i} f = \frac{\partial f}{\partial X_i}$ for all $1 \leq i \leq k$
- $\mathcal{D}_{X_1 X_2 - X_3^2} f = \frac{\partial^2 f}{\partial X_1 \partial X_2} - \frac{\partial^2 f}{\partial X_3^2}$

3 Higher-order vanishing of polynomials

3.1 Defining higher-order vanishing

Recall that we denote by $R_k^{(p)}$ the $\mathbb{F}_q$-vector space of homogeneous polynomials of degree p in $R_k = \mathbb{F}_q[X_1, \dots, X_k]$.

Definition 4. *Let $\boldsymbol{A} \in \mathbb{F}_q^{k\times n}$ and denote by $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n \in \mathbb{F}_q^k$ the columns of $\boldsymbol{A}$. We define the $\mathbb{F}_q$-linear map*

$$\varphi_{\boldsymbol{A}}^{(p)} \colon R_k^{(p)} \to \prod_{j=1}^{n} \mathbb{F}_q^{M_{k,p-2}}, \quad f \mapsto [\mathcal{D}_\nu f(\boldsymbol{a}_j)]_\nu^{j=1,\dots,n}$$

where ν runs through all monomials in k indeterminates of degree $0 \le d \le p-2$ and $M_{k,p-2} \coloneqq \sum_{d=0}^{p-2} \binom{k+d-1}{d}$ is the number of such monomials.

In other words, $\varphi_{\boldsymbol{A}}^{(p)}$ computes all partial derivatives of order at most $p-2$ of a given homogeneous polynomial $f \in R_k^{(p)}$ and evaluates them at all columns of $\boldsymbol{A}$. Note that for $p = 2$, the map $\varphi_{\boldsymbol{A}}^{(2)}$ simply evaluates homogeneous quadratic polynomials at all columns of $\boldsymbol{A}$. Hence the square distinguisher [21] essentially analyses the dimension of the kernel of $\varphi_{\boldsymbol{A}}^{(2)}$ in the cases where $\boldsymbol{A}$ is a random matrix or a parity check matrix of an alternant or Goppa code. For general p, the kernel of $\varphi_{\boldsymbol{A}}^{(p)}$ contains precisely those polynomials of degree p that simultaneously vanish of order $p-1$ at all points $\boldsymbol{a}_i$.

Remark 1. For any homogeneous polynomial $f \in R_k^{(u)}$ of degree u, it is easy to see that

$$\sum_{i=1}^{k} X_i \cdot \frac{\partial f}{\partial X_i} = u \cdot f$$

This is sometimes referred to as *Euler's theorem*. Suppose $\mathrm{char}(\mathbb{F}_q) > p$. The above identity implies that if all partial derivatives of order $p-2$ of $f \in R_k^{(p)}$ vanish at a given point, then so do all partial derivatives of order $0 \le d < p-2$. Thus, since we are mainly interested in the kernel, we may simplify the definition of $\varphi_{\boldsymbol{A}}^{(p)}$ by only computing the partial derivatives of order $p-2$ and evaluating them at all columns of $\boldsymbol{A}$. Similarly, in lower characteristic, some partial derivatives are superfluous, but others are not. For example, if $\mathrm{char}(\mathbb{F}_q) = 2$ and $f \in R_k^{(3)}$, then vanishing of first-order partial derivatives of f at a given point implies vanishing of f by Euler's theorem, but this need not be true e.g. for $f \in R_k^{(4)}$. We will analyse this more closely in section 4. For now, we remark that for concrete computations involving $\varphi_{\boldsymbol{A}}^{(p)}$, it makes sense to omit the computation of partial derivatives that are superfluous in the given characteristic.

Remark 2. Instead of using ordinary partial derivatives in Definition 4, one could also use *Hasse derivatives* [17]. These are a slightly adapted version of ordinary differentiation that have nicer properties in positive characteristic. For example, the fact that a polynomial is constant if and only if all its derivatives vanish is true for Hasse derivatives regardless of the characteristic of the ground field, but not for ordinary derivatives: For example, in characteristic 2, all derivatives of X^2 vanish, but X^2 is clearly not a constant polynomial. However, we stick to more familiar ordinary differentiation here, which yields equivalent results in the end.

We will analyse the dimension of the kernel of $\varphi_{\boldsymbol{A}}^{(p)}$ in the cases where $\boldsymbol{A}$ is the parity check matrix of a random alternant (or Goppa) code or of a random general linear code. Proposition 2 shows that this dimension depends only on the underlying code, but not on the choice of parity check matrix.

Proposition 2. *Suppose $\boldsymbol{A}, \boldsymbol{A'} \in \mathbb{F}_q^{k\times n}$ are row-equivalent. Then*

$$\dim_{\mathbb{F}_q}\left(\ker\left(\varphi_{\boldsymbol{A}}^{(p)}\right)\right) = \dim_{\mathbb{F}_q}\left(\ker\left(\varphi_{\boldsymbol{A'}}^{(p)}\right)\right)$$

Proof. Denoting by $\boldsymbol{X} = (X_1, \ldots, X_k)$ the row vector with indeterminates as entries, we claim that

$$\mathcal{D}_{\nu(\boldsymbol{X})}(f(\boldsymbol{X}\cdot\boldsymbol{S}^T)) = (\mathcal{D}_{\nu(\boldsymbol{X}\boldsymbol{S})}(f))(\boldsymbol{X}\cdot\boldsymbol{S}^T) \tag{1}$$

holds for any $\boldsymbol{S} \in \mathbb{F}_q^{k\times k}$ and $\nu, f \in R_k$. Note that since $\mathcal{D}_{\nu_1\cdot\nu_2} = \mathcal{D}_{\nu_1}\circ\mathcal{D}_{\nu_2}$ and $\mathcal{D}_{\alpha\nu_1+\beta\nu_2} = \alpha\cdot\mathcal{D}_{\nu_1} + \beta\cdot\mathcal{D}_{\nu_2}$ for all $\nu_1, \nu_2 \in R_k$ and $\alpha, \beta \in \mathbb{F}_q$, it suffices to prove the claim in the case that $\nu = X_j$ for some j. In this case, Equation 1 reduces to the ordinary chain rule for differentiation.

Letting $\boldsymbol{S}$ be an invertible matrix such that $\boldsymbol{A'} = \boldsymbol{S}\boldsymbol{A}$, we deduce from Equation 1 that the map

$$\ker\left(\varphi_{\boldsymbol{A'}}^{(p)}\right) \to \ker\left(\varphi_{\boldsymbol{A}}^{(p)}\right), \quad f(\boldsymbol{X}) \mapsto f(\boldsymbol{X}\boldsymbol{S}^T)$$

is well-defined, and it is an isomorphism since $\boldsymbol{S}$ is invertible. ☐

3.2 Definition of higher-order vanishing using vanishing ideals

An alternative way to define higher-order vanishing is via vanishing ideals. Given a point $\boldsymbol{c} \in \mathbb{F}_q^k$, we define the *vanishing ideal at* $\boldsymbol{c}$ by

$$\mathfrak{a}_{\boldsymbol{c}} = \{f \in R_k \mid f(\boldsymbol{c}) = 0\}$$

Given a matrix $\boldsymbol{A} \in \mathbb{F}_q^{k\times n}$ with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n \in \mathbb{F}_q^k$, we let

$$\phi_{\boldsymbol{A}} \colon R_k \to R_k/\mathfrak{a}_{\boldsymbol{a}_1}^{p-1} \times \cdots \times R_k/\mathfrak{a}_{\boldsymbol{a}_n}^{p-1}$$

be the map induced by the canonical projections. Note that the kernel of $\phi_{\boldsymbol{A}}$ is given by the ideal $\mathfrak{a}_{\boldsymbol{a}_1}^{p-1} \cap \ldots \cap \mathfrak{a}_{\boldsymbol{a}_n}^{p-1}$ which we call the *vanishing ideal of order* $p-1$ *at* $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$. Properties of this ideal have been studied in the literature, see for example [11]. The maps $\varphi_{\boldsymbol{A}}^{(p)}$ and $\phi_{\boldsymbol{A}}^{(p)}$ are closely related:

Proposition 3. *Suppose* char$(\mathbb{F}_q) > p$ *and let* $\boldsymbol{c} \in \mathbb{F}_q^k$. *The kernel of the map*

$$\varphi_{\boldsymbol{c}}^{(p)} \colon R_k^{(p)} \to \mathbb{F}_q^{\binom{k+p-3}{p-2}}, \quad f \mapsto [\mathcal{D}_\nu f(\boldsymbol{c})]_\nu$$

where ν *runs through all* $\binom{k+p-3}{p-2}$ *monomials of degree* $p-2$, *is given by* $\left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)}$.

Recall that by Remark 1, since char$(\mathbb{F}_q) > p$, it does not make a difference in the above statement whether we consider all partial derivatives of order $\leq p-2$ or only those of order equal to $p-2$.

Proof. The claim clearly holds if $\boldsymbol{c} = (0, \ldots, 0)$. So suppose $\boldsymbol{c} \neq (0, \ldots, 0)$. Without loss of generality, we may assume $c_1 = 1$.

The homogeneous linear polynomials $g_i = X_i - c_i X_1$ with $2 \leq i \leq k$ then generate the ideal $\mathfrak{a}_{\boldsymbol{c}}$. As a $\mathbb{F}_q$-vector space, $\left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)}$ is spanned by elements of the form $f g_{i_1} \cdots g_{i_{p-1}}$ where $2 \leq i_1, \ldots, i_{p-1} \leq k$ and $f \in R_k^{(1)}$. It follows directly from the product rule for differentiation and the fact that $g_i(\boldsymbol{c}) = 0$ for all i that any partial derivative of order $p-2$ of such an element vanishes at $\boldsymbol{c}$. This implies that $\left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)} \subset \ker\left(\varphi_{\boldsymbol{c}}^{(p)}\right)$.

We now prove $\ker\left(\varphi_{\boldsymbol{c}}^{(p)}\right) \subset \left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)}$ by induction on p. For $p = 2$, the claim follows immediately from the definition of $\mathfrak{a}_{\boldsymbol{c}}$. Now let $f \in \ker\left(\varphi_{\boldsymbol{c}}^{(p)}\right)$ for some $p > 2$. Then $\frac{\partial f}{\partial X_i} \in \ker\left(\varphi_{\boldsymbol{c}}^{(p-1)}\right)$ for all i, so by induction, $\frac{\partial f}{\partial X_i} \in \mathfrak{a}_{\boldsymbol{c}}^{p-2}$ for all i. By Euler's identity, $\sum_i X_i \cdot \frac{\partial f}{\partial X_i} = p \cdot f$, so since char$(\mathbb{F}_q) > p$, we deduce that $f \in \left(\mathfrak{a}_{\boldsymbol{c}}^{p-2}\right)^{(p)}$. Thus we can write

$$f = h_1 + X_1 \cdot h_2 + X_1^2 \cdot \sum_{2 \leq i_1 \leq \ldots \leq i_{p-2} \leq k} \lambda_{i_1, \ldots, i_{p-2}} \cdot g_{i_1} \cdots g_{i_{p-2}}$$

for some $h_1 \in (\mathfrak{a}_{\boldsymbol{c}}^p)^{(p)}$ and $h_2 \in \left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p-1)}$ and $\lambda_{i_1, \ldots, i_{p-2}} \in \mathbb{F}_q$. Then for any $2 \leq i_1 \leq \ldots \leq i_{p-2} \leq k$, we have

$$\frac{\partial^{p-2} f}{\partial X_{i_1} \cdots \partial X_{i_{p-2}}}(\boldsymbol{c}) = \lambda_{i_1, \ldots, i_{p-2}} \cdot \mu_2! \mu_3! \cdots \mu_k!$$

where μ_i is the number of elements among $i_1, \ldots, i_{p-2}$ that are equal to i. This must vanish because $f \in \ker\left(\varphi_{\boldsymbol{c}}^{(p)}\right)$. Since char$(\mathbb{F}_q) > p$, this implies that $\lambda_{i_1, \ldots, i_{p-2}} = 0$ for all $2 \leq i_1 \leq \ldots \leq i_{p-2} \leq k$ and hence $f \in \left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)}$ as claimed. $\square$

Proposition 3 implies that in large characteristic and for any matrix $\boldsymbol{A}$, the maps $\varphi_{\boldsymbol{A}}^{(p)}$ and $\phi_{\boldsymbol{A}}^{(p)}$ have the same kernel. In lower characteristic, one has to be a bit more careful. For example if char$(\mathbb{F}_q) = 2$, the polynomial $X_1^{p-2} \cdot (X_2 - c_2 X_1)^2$ lies in the kernel of $\varphi_{\boldsymbol{A}}^{(p)}$ for any $p \geq 2$. But it is not an element of $\left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)}$ for

$p \geq 4$. However, following an argument similar to the proof of Proposition 3, one can show the following: If we restrict $\varphi_{\boldsymbol{A}}^{(p)}$ to the subspace $U_k^{(p)}$ of homogeneous polynomials of degree p where each indeterminate occurs to degree at most 1, its kernel is equal to $\left(\mathfrak{a}_{\boldsymbol{c}}^{p-1}\right)^{(p)} \cap U_k^{(p)}$.

We will analyse the case of characteristic 2 more closely further below. In the following, we will only work with the definition of higher-order vanishing via differentials, i.e. with $\varphi_{\boldsymbol{A}}^{(p)}$. However, we will make use of ideas from works studying the higher-order vanishing ideal.

3.3 Reduction to multilinear polynomials

By Proposition 2, the dimension of the kernel of $\varphi_{\boldsymbol{A}}^{(p)}$ is invariant under row operations on the matrix $\boldsymbol{A}$. Let $\boldsymbol{A}'$ be the (unique) reduced row-echelon form of $\boldsymbol{A}$. Assuming $\boldsymbol{A}$ has full rank k, there are k pivot columns $\boldsymbol{a}_{i_1}, \ldots, \boldsymbol{a}_{i_k}$ of $\boldsymbol{A}'$ that are equal to the standard basis vectors in $\mathbb{F}_q^k$. Lemma 1 below determines the polynomials of higher-order vanishing at the standard basis vectors. First, we need a definition.

Definition 5. *We define*

$$U_k^{(p)} = \mathit{span}_{\mathbb{F}_q}\left\{X_1^{d_1}\cdots X_k^{d_k} \mid d_i \leq 1 \forall i \quad \mathit{and} \quad \sum_{i=1}^{k} d_i = p\right\}$$

to be the subspace spanned by monomials in $R_k^{(p)}$ where each indeterminate occurs to degree at most 1.

Lemma 1. *Let $\boldsymbol{I}_k$ denote the $k \times k$ identity matrix over $\mathbb{F}_q$ and $c := \mathit{char}(\mathbb{F}_q)$. For any $p \geq 2$, we have*

$$\ker\left(\varphi_{\boldsymbol{I}_k}^{(p)}\right) = U_k^{(p)} \oplus \langle X_i^c X_j^c \mid 1 \leq i < j \leq k\rangle^{(p)}$$

where $\langle\cdot\rangle$ denotes the ideal in R_k generated by the indicated monomials.

Proof. It is clear that $U_k^{(p)} \cap \langle X_i^c X_j^c \mid 1 \leq i < j \leq k\rangle^{(p)} = \{0\}$ and easy to see that $U_k^{(p)} \oplus \langle X_i^c X_j^c \mid 1 \leq i < j \leq k\rangle^{(p)} \subset \ker\left(\varphi_{\boldsymbol{I}_k}^{(p)}\right)$.

Now suppose that $f \in \ker\left(\varphi_{\boldsymbol{I}_k}^{(p)}\right)$. Consider a monomial of the form

$$X_{i_0}^{d_0} X_{i_1}^{d_1} \cdots X_{i_\ell}^{d_\ell} \in R_k^{(p)}$$

with $i_0, \ldots, i_\ell$ pairwise distinct, $d_0 \geq 2$ and $d_1, \ldots, d_\ell < c$. We want to show that the coefficient of any such monomial in f is zero. Suppose this is not the case and consider a monomial as above with coefficient $\lambda \neq 0$ in f. Then we have

$$0 = \frac{\partial^{d_1+\ldots+d_\ell} f}{\partial X_{i_1}^{d_1}\cdots \partial X_{i_\ell}^{d_\ell}}(\boldsymbol{e}_{i_0}) = \lambda \cdot d_1! \cdots d_\ell!$$

where we used the fact that $f \in \ker\left(\varphi^{(p)}_{\boldsymbol{I}_k}\right)$ for the first equality. Since $d_1, \dots, d_\ell$ are bounded by c, this implies that $\lambda = 0$, which is a contradiction. Hence our claim follows. But then it follows directly that indeed,

$$f \in U_k^{(p)} \oplus \langle X_i^c X_j^c \mid 1 \leq i < j \leq k \rangle^{(p)}$$

$\square$

We clearly have $\ker\left(\varphi^{(p)}_{\boldsymbol{A}'}\right) \subset \ker\left(\varphi^{(p)}_{\boldsymbol{I}_k}\right)$. The space $\langle X_i^c X_j^c \mid 1 \leq i < j \leq k \rangle^{(p)}$ consists of elements that lie in the kernel purely due to the characteristic of our ground field, irrespective of the value of $\boldsymbol{A}'$ and its entries. Thus, for our distinguisher, we are only interested in the part of the kernel that lies in $U_k^{(p)}$. Note that if we differentiate any polynomial in $U_k^{(p)}$ by some variable more than once, this yields the zero polynomial. Hence it suffices to consider only partial derivatives that differentiate by any given variable at most once. This motivates the following slightly adapted definition. It reduces the size of the domain and range compared to $\varphi^{(p)}_{\boldsymbol{A}}$, but still captures the interesting part of its kernel.

Definition 6. *Let $\boldsymbol{A} \in \mathbb{F}_q^{k \times n}$ be a matrix of rank k. Let $\tilde{\boldsymbol{A}} \in \mathbb{F}_q^{k \times (n-k)}$ be the matrix obtained from $\boldsymbol{A}$ by first computing the (unique) reduced row-echelon form of $\boldsymbol{A}$ and then removing the k columns that contain the pivot entries, i.e. that contain the leading 1 in a row. Let $\tilde{\boldsymbol{a}}_1, \dots, \tilde{\boldsymbol{a}}_{n-k}$ denote the columns of $\tilde{\boldsymbol{A}}$. We define*

$$\tilde{\varphi}^{(p)}_{\boldsymbol{A}} : U_k^{(p)} \to \prod_{j=1}^{n-k} \mathbb{F}_q^{\tilde{M}_{k,p-2}}, \quad f \mapsto [\mathcal{D}_\nu f(\tilde{\boldsymbol{a}}_j)]_\nu^{j=1,\dots,n-k}$$

where ν runs through all monomials of degrees $0 \leq d \leq p-2$ where each indeterminate has degree at most 1, and $\tilde{M}_{k,p-2} \coloneqq \sum_{d=0}^{p-2} \binom{k}{d}$ is the number of such monomials.

Recall that as pointed out in Remark 1, depending on the characteristic of the ground field, for computations we may omit partial derivatives of certain orders in the definition above.

4 Behaviour of higher-order vanishing at randomly selected points

In order to analyse the rank of $\tilde{\varphi}^{(p)}_{\boldsymbol{A}}$ for a randomly selected matrix $\boldsymbol{A} \in \mathbb{F}_q^{k \times n}$ of rank k, we show that it can be interpreted as the Hilbert function in degree p of a certain ideal of polynomials. Results in this direction have already been described in the literature. [11, Theorem I] establishes a connection between higher-order vanishing ideals and ideals of certain polynomials that we also consider in the following, using what the authors call *inverse systems of ideals*. This is basically the content of Proposition 4 below. We shall give a self-contained exposition of our results in this section, but remark that it is heavily inspired by this work.

Since we only consider polynomials in $U_k^{(p)}$, it will be convenient to work in the quotient ring $\tilde{R}_k \coloneqq R_k/\langle X_1^2, \ldots, X_k^2\rangle$. Note that since we quotient by a homogeneous ideal, this inherits a grading from R_k.

4.1 Interpreting higher-order vanishing in terms of polynomial ideals

The following polynomials are central to making the link between our map φ defined in terms of differentials and polynomial ideals.

Definition 7. *For $\boldsymbol{c} = (c_1, \ldots, c_k) \in \mathbb{F}_q^k$ and every integer $a \geq 1$ we define the polynomial*

$$L_{\boldsymbol{c}}^{[a]} \coloneqq \sum_{1 \leq i_1 < \ldots < i_a \leq k} c_{i_1} \cdots c_{i_a} X_{i_1} \cdots X_{i_a} \in \tilde{R}_k^{(a)}$$

These polynomials are multiplicatively related in the following way:

Lemma 2. *For any $\boldsymbol{c} \in \mathbb{F}_q^k$ and $a, b \geq 1$, the following identity holds in $\tilde{R}_k$:*

$$L_{\boldsymbol{c}}^{[a]} \cdot L_{\boldsymbol{c}}^{[b]} = \binom{a+b}{a} \cdot L_{\boldsymbol{c}}^{[a+b]}$$

Proof. Note that all monomials in the product $L_{\boldsymbol{c}}^{[a]} \cdot L_{\boldsymbol{c}}^{[b]}$ that contain an indeterminate of degree greater than 1 vanish in $\tilde{R}_k$. Any monomial of degree $a+b$ such that every indeterminate is of degree at most 1 occurs as many times as there are partitions of the set of occurring indeterminates in a set of size a and a set of size b. There are $\binom{a+b}{a}$ such partitions. This implies the claim. □

The following simple observation is central because it allows us to rephrase evaluation of a homogeneous polynomial at a point (occurring in the definition of $\varphi_{\boldsymbol{A}}^{(p)}$) as differentiation with respect to a polynomial as defined in Definition 7.

Lemma 3. *For any $\boldsymbol{c} \in \mathbb{F}_q^k$ and $\nu \in \tilde{R}_k^{(p-a)}$ and $f \in U_k^{(p)}$, the following holds:*

$$\mathcal{D}_{\nu \cdot L_{\boldsymbol{c}}^{[a]}} f = (\mathcal{D}_\nu f)(\boldsymbol{c}) \tag{2}$$

Proof. Note that the above expressions are in fact well-defined: Since $f \in U_k^{(p)}$, we have $\mathcal{D}_{X_i^2} f = 0$ for all i and hence the expression $\mathcal{D}_\mu f$ is well-defined for $\mu \in \tilde{R}_k$ (and not only for $\mu \in R_k$). Furthermore, note that the expression on the left in Equation 2 is strictly speaking an element in $R_k^{(0)}$, which we identify with $\mathbb{F}_q$, and the quantity to the right of the equality is an element of $\mathbb{F}_q$ by definition.

By linearity, it suffices to prove the statement for the case that ν and f are monomials. Since $\mathcal{D}_{\nu \cdot L_{\boldsymbol{c}}^{[a]}} f = \mathcal{D}_{L_{\boldsymbol{c}}^{[a]}}(\mathcal{D}_\nu(f))$, it suffices to prove that $\mathcal{D}_{L_{\boldsymbol{c}}^{[a]}}(g) = g(\boldsymbol{c})$ for any $g \in U_k^{(a)}$. But this is immediate from the definition of $L_{\boldsymbol{c}}^{[a]}$ since g is square-free. □

Now let $\boldsymbol{A} \in \mathbb{F}_q^{k \times n}$ have rank k and let $\tilde{\boldsymbol{A}} \in \mathbb{F}_q^{k \times \tilde{n}}$ with $\tilde{n} := n - k$ be the matrix obtained by transforming $\boldsymbol{A}$ to reduced row-echelon form and omitting the pivot columns. Let $\tilde{\boldsymbol{a}}_1, \ldots, \tilde{\boldsymbol{a}}_{\tilde{n}} \in \mathbb{F}_q^k$ denote the columns of $\tilde{\boldsymbol{A}}$. Note that Lemma 3 implies that the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ is trivial if and only if

$$\langle L_{\tilde{\boldsymbol{a}}_i}^{[j]} \mid 1 \leq i \leq \tilde{n},\ 2 \leq j \leq p \rangle^{(p)} = \tilde{R}_k^{(p)}$$

Indeed, one can show more:

Proposition 4. *With notation as above,*

$$\dim_{\mathbb{F}_q} \left(\langle L_{\tilde{\boldsymbol{a}}_i}^{[j]} \mid 1 \leq i \leq \tilde{n},\ 2 \leq j \leq p \rangle^{(p)} \right) = \dim_{\mathbb{F}_q} \left(Im \left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)} \right) \right) \tag{3}$$

where $\langle L_{\tilde{\boldsymbol{a}}_i}^{[j]} \mid 1 \leq i \leq \tilde{n},\ 2 \leq j \leq p \rangle$ *denotes the ideal in* $\tilde{R}_k$ *generated by the given polynomials.*

Proof. We pick some ordering of monomials in $\tilde{R}_k$ of degree p and consider the degree p Macaulay matrix $\boldsymbol{M}$ associated with the $L_{\tilde{\boldsymbol{a}}_i}^{[j]}$. That is, each row of $\boldsymbol{M}$ corresponds to a product $\nu \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j]}$ for some i, j and a monomial ν of degree $p - j$, and the entries in that row contain the $\mathbb{F}_q$-coefficients of the monomials in $\nu \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j]}$ (in the specified order). By construction, the rank of $\boldsymbol{M}$ is the expression on the left of Equation 3.

On the other hand, consider the matrix $\boldsymbol{P}$ representing the linear map $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ with respect to the basis of $U_k^{(p)}$ consisting of square-free monomials of degree p with the same ordering as above. Now notice that by Lemma 3, the row of $\boldsymbol{P}$ corresponding to the differential $\mathcal{D}_\nu$ for some ν of degree $p - j$ and evaluation at $\tilde{\boldsymbol{a}}_i$ is *equal* to the row of $\boldsymbol{M}$ corresponding to the product $\nu \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j]}$. Hence $\boldsymbol{P}$ and $\boldsymbol{M}$ have the same rank. This implies the claim. $\square$

By Lemma 2, depending on the characteristic of the ground field, some of the generators of the ideal $\langle L_{\tilde{\boldsymbol{a}}_i}^{[j]} \mid 1 \leq i \leq \tilde{n},\ 2 \leq j \leq p \rangle$ are superfluous:

Corollary 1. *Let* $c := char(\mathbb{F}_q)$ *and* $\boldsymbol{A}, \tilde{\boldsymbol{A}}$ *be as above. Then*

$$\dim_{\mathbb{F}_q} \left(\langle L_{\tilde{\boldsymbol{a}}_i}^{[2]}, L_{\tilde{\boldsymbol{a}}_i}^{[c^\ell]} \mid 1 \leq i \leq \tilde{n},\ 1 \leq \ell \leq \lfloor \log_c(p) \rfloor \rangle^{(p)} \right) = \dim_{\mathbb{F}_q} \left(Im \left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)} \right) \right)$$

Proof. Consider an integer j such that $2 < j < c$. (In the case $c = 2$ or $c = 3$, this does not occur of course.) By Lemma 2 we have

$$L_{\tilde{\boldsymbol{a}}_i}^{[2]} \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j-2]} = \binom{j}{2} \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j]}$$

Since $2 < j < c$, the binomial coefficient is non-zero in $\mathbb{F}_q$. Hence $L_{\tilde{\boldsymbol{a}}_i}^{[j]} \in \langle L_{\tilde{\boldsymbol{a}}_i}^{[2]}, L_{\tilde{\boldsymbol{a}}_i}^{[c^\ell]} \mid 1 \leq i \leq \tilde{n},\ 1 \leq \ell \leq \lfloor \log_c(p) \rfloor \rangle$.

Now suppose $j > c$ is not a power of c. Let c^ℓ be the largest power of c less than j. By Lemma 2 we have

$$L_{\tilde{\boldsymbol{a}}_i}^{[c^\ell]} \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j-c^\ell]} = \binom{j}{c^\ell} \cdot L_{\tilde{\boldsymbol{a}}_i}^{[j]}$$

Lucas's theorem [7, Theorem 4.71] implies that $\binom{j}{c^\ell}$ is not divisible by c. Hence $L_{\tilde{\boldsymbol{a}}_i}^{[j]} \in \langle L_{\tilde{\boldsymbol{a}}_i}^{[2]}, L_{\tilde{\boldsymbol{a}}_i}^{[c^\ell]} \mid 1 \leq i \leq \tilde{n},\ 1 \leq \ell \leq \lfloor \log_c(p) \rfloor \rangle$. The claim now follows from Proposition 4. $\square$

We have reformulated the rank (or, equivalently, the dimension of the kernel) of the map $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ in terms of the dimension of the degree p component of a polynomial ideal. This yields an equivalent definition of our distinguisher that does not make reference to differentiation. It turns out (see the next section) that the analysis of the case of random matrices $\boldsymbol{A}$ is easier using this formulation.

4.2 Semi-regularity and behaviour in the random case

Let $\boldsymbol{A} \in \mathbb{F}_q^{k \times n}$ be a randomly selected matrix of rank k. We want to study the dimension of the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$. Note that $\tilde{\boldsymbol{A}}$ (obtained by bringing $\boldsymbol{A}$ into reduced row-echelon form and removing the pivot columns) is then a randomly selected $k \times \tilde{n}$ matrix where $\tilde{n} \coloneqq n - k$. Denoting by $\tilde{\boldsymbol{a}}_1, \ldots, \tilde{\boldsymbol{a}}_{\tilde{n}}$ the columns of $\tilde{\boldsymbol{A}}$ and setting

$$\mathfrak{L}_{\tilde{\boldsymbol{A}}} \coloneqq \langle L_{\tilde{\boldsymbol{a}}_i}^{[j]} \mid 1 \leq i \leq \tilde{n},\ j \geq 2 \rangle \tag{4}$$

to be the ideal in $\tilde{R}_k$ we considered before, Proposition 4 implies that

$$\dim_{\mathbb{F}_q}\left(\ker\left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)}\right)\right) = \dim_{\mathbb{F}_q}\left(\tilde{R}_k^{(p)} / \mathfrak{L}_{\tilde{\boldsymbol{A}}}^{(p)}\right) \tag{5}$$

This dimension can be explicitly computed in the case that polynomials generating the ideal $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ form a so-called *semi-regular sequence*. Intuitively, this means the polynomials have as few relations between them as possible. We recall:

Definition 8. *Let $\tilde{R}_k = \mathbb{F}_q[X_1, \ldots, X_k]/\langle X_1^2, \ldots, X_k^2 \rangle$. Let d be a positive integer. A sequence $f_1, \ldots, f_n \in \tilde{R}_k$ of homogeneous elements is* d-regular *if the following holds, depending on char$(\mathbb{F}_q)$:*

- *If char$(\mathbb{F}_q) \neq 2$: For any $1 \leq i \leq n$, if $g \in \tilde{R}_k$ such that $gf_i \in \langle f_1, \ldots, f_{i-1} \rangle$ and $\deg(gf_i) < d$, then $g \in \langle f_1, \ldots, f_{i-1} \rangle$.*
- *If char$(\mathbb{F}_q) = 2$: For any $1 \leq i \leq n$, if $g \in \tilde{R}_k$ such that $gf_i \in \langle f_1, \ldots, f_{i-1} \rangle$ and $\deg(gf_i) < d$, then $g \in \langle f_1, \ldots, f_{i-1}, f_i \rangle$.*

We define the degree of regularity *of $I = \langle f_1, \ldots, f_n \rangle$ as*

$$d_{reg}(I) \coloneqq \min\left\{ d \geq 0 \mid \dim_{\mathbb{F}_q}(I^{(d)}) = \binom{k}{d} \right\}$$

where $I^{(d)}$ denotes the subspace of elements of degree d in I. The sequence $f_1, \ldots, f_n$ is semi-regular *if it is d_{reg}-regular.*

Remark 3. The above definition of semi-regularity for characteristic 2 agrees with [4, Definition 9] which, like our definition, defines semi-regularity for elements in $\tilde{R}_k$. For characteristic not equal to 2, [4, Definition 5] defines semi-regularity for elements in R_k while we defined it for elements in $\tilde{R}_k$. Note that our definition is equivalent to the semi-regularity of the sequence

$$X_1^2, \ldots, X_k^2, \hat{f}_1, \ldots, \hat{f}_n \in R_k$$

according to [4, Definition 5], where $\hat{f}_i$ is a lift of f_i to R_k.

For $(d+1)$-regular sequences, one can easily determine the Hilbert series up to degree d. Let us introduce some useful notation: Given a formal power series $S = \sum_{i=0}^{\infty} a_i t^i$ with integer coefficients and letting $d \in \mathbb{N}_0 \cup \{\infty\}$ denote the smallest index such that $a_d < 0$, we define the power series

$$[S]^+ \coloneqq \sum_{i=0}^{\infty} a_i' \cdot t^i \quad \text{where } a_i' = \begin{cases} a_i & \text{if } i < d \\ 0 & \text{otherwise} \end{cases}$$

Proposition 5. *Let $f_1, \ldots, f_n \in \tilde{R}_k$ be homogeneous with $\deg(f_i) = d_i > 0$ for all i such that they form a semi-regular sequence, and set $I \coloneqq \langle f_1, \ldots, f_n \rangle$. We define the power series*

$$S_{n,k}^{\boldsymbol{d}} = (1+t)^k \cdot \prod_{i=1}^{n} (1 - t^{d_i}), \quad T_{n,k}^{\boldsymbol{d}} = \frac{(1+t)^k}{\prod_{i=1}^{n}(1+t^{d_i})}$$

Then the Hilbert series $HS_I(t) = \sum_{i \geq 0} \dim_{\mathbb{F}_q}(\tilde{R}_k^{(i)}/I^{(i)}) \cdot t^i$ of I is given by $[S_{n,k}^{\boldsymbol{d}}]^+$ if char$(\mathbb{F}_q) \neq 2$ and by $[T_{n,k}^{\boldsymbol{d}}]^+$ if char$(\mathbb{F}_q) = 2$.

Proof. This follows immediately from [4, Proposition 6] if $\mathrm{char}(\mathbb{F}_q) \neq 2$ and from [4, Proposition 10] otherwise. ☐

Remark 4. It is conjectured that if the length of a sequence of polynomials is sufficiently large compared to the number of variables, then most such random sequences are semi-regular. However, little has been proven in this direction. [15] and [13] characterise the conditions under which a sequence of *one* or *two* quadratic polynomials in $\mathbb{F}_2[X_1, \ldots, X_k] / \langle X_1^2, \ldots, X_k^2 \rangle$ is semi-regular. [13, Theorem 4.4] shows that if the length of a sequence is too small (namely at most $k/4 - 1$), it cannot be semi-regular. On the other hand, [24, Theorem 1.1] gives a lower bound on the proportion of sequences that are semi-regular if the length of the sequence is at least $\lceil (k-1)(k-2)/6 \rceil$, and the bound tends to 1 as $k \to \infty$. In between, not much has been rigorously proven. See also the introduction of [13] for a discussion.

Let us now consider the polynomials $L_{\tilde{a}_i}^{[j]}$ and the ideal $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$. The polynomials $L_{\tilde{a}_i}^{[j]}$ are of course not chosen uniformly at random, even in our case that the matrix $\tilde{\boldsymbol{A}}$ has been chosen uniformly at random. Furthermore, we saw in Lemma 2 that there are multiplicative relations between them so that the full set of them cannot form a semi-regular sequence. Corollary 1 provides a smaller set of generators. One may hope that, having excluded the trivial relations, these now form a semi-regular sequence. However, we need to be a bit careful about the characteristic of the ground field:

Example 1. Suppose $\mathrm{char}(\mathbb{F}_q) = 3$. Note that in this case, by Lemma 2 we have $L_{\boldsymbol{c}}^{[1]} \cdot L_{\boldsymbol{c}}^{[2]} = 0$ for any $\boldsymbol{c} \in \mathbb{F}_q^k$. Hence a sequence consisting of a single polynomial $L_{\boldsymbol{c}}^{[2]}$ is not 4-regular. Thus, in this case, the sequence of polynomials described in Corollary 1 is generally not semi-regular.

For now, let us restrict to the cases of characteristic 2 and of characteristic $> p$. Our experiments indeed suggest that if $\tilde{\boldsymbol{A}}$ has been chosen uniformly at random and parameter sizes are sufficiently large, then with high probability, the sequences of polynomials $L_{\tilde{\boldsymbol{a}}_i}^{[j]}$ (excluding the superfluous ones) indeed form a semi-regular sequence if $\mathrm{char}(\mathbb{F}_q) = 2$, and they are at least regular up to degree p if $\mathrm{char}(\mathbb{F}_q) > p$. See Remark 5 for some discussion and experimental evidence. We summarise this in the following heuristic assumptions:

Heuristic Assumption 1. Suppose $\mathrm{char}(\mathbb{F}_q) = 2$ and $\tilde{n} > k$ and let $\tilde{\boldsymbol{A}} \in \mathbb{F}_q^{k \times \tilde{n}}$ be a randomly selected matrix with columns $\tilde{\boldsymbol{a}}_1, \ldots, \tilde{\boldsymbol{a}}_{\tilde{n}}$. If k is sufficiently large, then the polynomials $L_{\tilde{\boldsymbol{a}}_i}^{[2^\ell]}$ for $\ell \geq 1$ and $i = 1, \ldots, \tilde{n}$ form a semi-regular sequence in $\tilde{R}_k$ with high probability.

Heuristic Assumption 2. Let $p \geq 2$ and suppose $\mathrm{char}(\mathbb{F}_q) > p$ and $\tilde{n} > k$. Let $\tilde{\boldsymbol{A}} \in \mathbb{F}_q^{k \times \tilde{n}}$ be a randomly selected matrix with columns $\tilde{\boldsymbol{a}}_1, \ldots, \tilde{\boldsymbol{a}}_{\tilde{n}}$. If k is sufficiently large, then the polynomials $L_{\tilde{\boldsymbol{a}}_i}^{[2]}$ for $i = 1, \ldots, \tilde{n}$ form a $(p+1)$-regular sequence in $\tilde{R}_k$ with high probability.

Now, assuming Heuristic Assumption 1 and Heuristic Assumption 2 hold, we are able to predict (at least with high probability for sufficiently large parameter sets) the dimension of the kernel of our map $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ for randomly chosen $\boldsymbol{A}$. As already indicated above, these formulas have been confirmed in computations for concrete parameters, see also Remark 5.

Corollary 2. *Suppose Heuristic Assumption 1 and Heuristic Assumption 2 hold and $\boldsymbol{A} \in \mathbb{F}_q^{k \times n}$ is randomly selected. Let $\tilde{n} \coloneqq n - k > k$. If k is sufficiently large, then the following holds with high probability: If $\mathrm{char}(\mathbb{F}_q) = 2$ or $\mathrm{char}(\mathbb{F}_q) > p$, then*

$$\begin{aligned}\dim_{\mathbb{F}_q}\left(\ker\left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)}\right)\right) &= \textit{degree } p \textit{ coefficient of } \left[(1+t)^k \cdot \left(1-t^2\right)^{\tilde{n}}\right]^+ \\ &= \begin{cases} \sum_{i=0}^{\lfloor p/2 \rfloor} (-1)^i \cdot \binom{\tilde{n}}{i} \cdot \binom{k}{p-2i} & \textit{if } p < d_{reg} \\ 0 \quad \textit{if } p \geq d_{reg} \end{cases}\end{aligned}$$

Proof. For $\mathrm{char}(\mathbb{F}_q) > p$, this follows immediately from Equation 5, Corollary 1 and Proposition 5. If $\mathrm{char}(\mathbb{F}_q) = 2$, Proposition 5 yields that

$$\dim_{\mathbb{F}_q}\left(\ker\left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)}\right)\right) = \text{degree } p \text{ coefficient of } \left[\frac{(1+t)^k}{\prod_{\ell=1}^{\infty}(1+t^{2^\ell})^{\tilde{n}}}\right]^+$$

Note that $\frac{1}{\prod_{\ell=1}^{\infty}(1+t^{2^\ell})} = 1 - t^2$, so the result follows. ☐

We see that under the above heuristic assumptions, we obtain the same formula in the cases $\mathrm{char}(\mathbb{F}_q) = 2$ and $\mathrm{char}(\mathbb{F}_q) > p$. This is not a coincidence. Consider the generators $L_{\tilde{\boldsymbol{a}}_i}^{[\ell]}$ of the ideal $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ where $\ell \geq 2$. The essential difference between different characteristics of the ground field is (or at least seems to be) that by Lemma 2, depending on the characteristic, $L_{\boldsymbol{c}}^{[i]} \cdot L_{\boldsymbol{c}}^{[j]}$ is either 0 or a non-zero multiple of $L_{\boldsymbol{c}}^{[i+j]}$. However, since we include all $L_{\boldsymbol{c}}^{[\ell]}$ in the ideal $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ anyway, this should not make a difference for the vector space dimension of $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ in degree p. Note that this suggests that the formula from Corollary 2 should actually hold for *all* characteristics, even though we cannot deduce it from semi-regularity assumptions in the case that $2 < \mathrm{char}(\mathbb{F}_q) < p$ (see Example 1). Indeed, this is supported by experimental evidence (see Remark 5). We summarise this in the following heuristic which we shall use to predict the behaviour of our distinguisher in the random case:

Heuristic 1. Suppose $\boldsymbol{A} \in \mathbb{F}_q^{k\times n}$ is randomly selected. Let $\tilde{n} := n - k > k$. If k is sufficiently large, then with high probability,

$$\dim_{\mathbb{F}_q}\left(\ker\left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)}\right)\right) = \text{degree } p \text{ coefficient of } \left[(1+t)^k\left(1-t^2\right)^{\tilde{n}}\right]^+$$

Table 1. Results of computing the Hilbert series of $\tilde{R}_k/\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ for randomly chosen $\tilde{\boldsymbol{A}} \in \mathbb{F}_q^{k\times\tilde{n}}$ for $\tilde{n} = 4$ and $k = 12$ for different values of q. For $q = 2$, we ran the experiment 100,000 times, for $q = 16$ and $q = 9$ we ran it 100 times.

p	0	1	2	3	4	5
Coefficient of $\left[(1+t)^k \cdot \left(1-t^2\right)^{\tilde{n}}\right]^+$ in degree p	1	12	62	172	237	0
Percentage of correct results for $q = 2$	100%	100%	98.5%	64.7%	15.4%	0%
Percentage of correct results for $q = 16$	100%	100%	100%	100%	100%	98%
Percentage of correct results for $q = 9$	100%	100%	100%	100%	100%	78%

Remark 5. Note that we are being vague about what constitutes 'sufficiently large' parameters. Indeed, if k as well as the field size q and $\tilde{n}$ are quite small (say $k = 12$ and $q = 2$ and $\tilde{n} = 4$), the dimensions one obtains are close to the predictions above, but can randomly fluctuate. If the field size is chosen to be larger (say $q = 16$), then even for such small k and $\tilde{n}$, the dimensions behave

as predicted in Heuristic 1, which also means the sequences of polynomials are indeed semi-regular. See Table 1 for some concrete experimental results that illustrate this. Of course, interesting parameter sets have much larger k and $\tilde{n}$. In particular, $\tilde{n}$ is usually significantly larger than k. See, for example, the Classic McEliece parameter sets in subsection 6.2 or those considered in subsection 6.1. Thus, for interesting parameter sets, we observe that the dimensions are precisely predicted by Heuristic 1 also in the case that $q = 2$.

5 Higher-order vanishing for (shortened) alternant and Goppa codes

Suppose $\boldsymbol{A} \in \mathbb{F}_q^{mr \times n}$ is a parity check matrix of an alternant code $\mathrm{Alt}_r(\boldsymbol{x}, \boldsymbol{y})$ over $\mathbb{F}_q$ where $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{F}_{q^m}^n$. We assume throughout that the alternant code is proper, i.e. that $\boldsymbol{A}$ has full rank mr.

Since $\varphi_{\boldsymbol{A}}^{(p)}$ is $\mathbb{F}_q$-linear, we may extend scalars to $\mathbb{F}_{q^m}$. This preserves the dimension of its kernel. By Proposition 1, over the extension field $\mathbb{F}_{q^m}$, the parity check matrix $\boldsymbol{A}$ is row-equivalent to the matrix

$$\boldsymbol{A}_{\mathbb{F}_{q^m}} = \begin{pmatrix} \boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}} \\ (\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}})^{(q)} \\ (\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}})^{(q^2)} \\ \cdots \\ (\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}})^{(q^{m-1})} \end{pmatrix} \text{ with } \boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}} = \begin{pmatrix} y_1 & y_2 & \dots & y_n \\ y_1x_1 & y_2x_2 & \dots & y_nx_n \\ y_1x_1^2 & y_2x_2^2 & \dots & y_nx_n^2 \\ \dots & \dots & \dots & \dots \\ y_1x_1^{r-1} & y_2x_2^{r-1} & \dots & y_nx_n^{r-1} \end{pmatrix}$$

where $(\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}})^{(q^i)}$ denotes the entry-wise q^i-th power of the matrix $\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}}$. By Proposition 2, the dimension of the kernel of the $\mathbb{F}_q$-linear map $\varphi_{\boldsymbol{A}}^{(p)}$ and the $\mathbb{F}_{q^m}$-linear map $\varphi_{\boldsymbol{A}_{\mathbb{F}_{q^m}}}^{(p)}$ are equal.

It is now convenient to label the indeterminates in R_{rm} as

$$X_0^{(0)}, \dots, X_{r-1}^{(0)}, X_0^{(1)}, \dots, X_{r-1}^{(1)}, \dots, X_0^{(m-1)}, \dots, X_{r-1}^{(m-1)}$$

corresponding to the sub-matrices $(\boldsymbol{G}_{\boldsymbol{x},\boldsymbol{y}})^{(q^i)}$ of $\boldsymbol{A}_{\mathbb{F}_{q^m}}$. Now observe that $\varphi_{\boldsymbol{A}_{\mathbb{F}_{q^m}}}^{(p)}$ factors through the map

$$\psi_{m,r}^{(p)} \colon \mathbb{F}_{q^m}\left[X_i^{(j)}\right]^{(p)} \to \left(\mathbb{F}_{q^m}\left[X, Y\right]\right)^{M_{mr,p-2}}, \quad f \mapsto \left[\mathcal{D}_\nu(f)|_{X_i^{(j)} = (YX^i)^{q^j}}\right]_\nu$$

where $M_{k,p-2} = \sum_{d=0}^{p-2} \binom{k+d-1}{d}$, $i = 0, \dots, r-1$ and $j = 0, \dots, m-1$ and ν runs through all monomials in $R_{rm} = \mathbb{F}_{q^m}\left[X_i^{(j)}\right]$ of degree at most $p-2$.

Thus the kernel of $\psi_{m,r}^{(p)}$ is contained in the kernel of $\varphi_{\boldsymbol{A}_{\mathbb{F}_{q^m}}}^{(p)}$.

5.1 The kernel in the alternant case

Let $e = \left\lfloor \log_q \left(\frac{r-1}{p-1}\right) \right\rfloor$. For $0 \le u \le e$ and $0 \le v < m$, we define the $p \times (r-(p-1)q^u)$ matrix

$$\boldsymbol{B}_{u,v} \coloneqq \begin{pmatrix} X_0^{(v)} & X_1^{(v)} & \dots & X_{r-1-(p-1)q^u}^{(v)} \\ X_{q^u}^{(v)} & X_{1+q^u}^{(v)} & \dots & X_{r-1-(p-2)q^u}^{(v)} \\ \dots & \dots & \dots & \dots \\ X_{(p-1)q^u}^{(v)} & X_{1+(p-1)q^u}^{(v)} & \dots & X_{r-1}^{(v)} \end{pmatrix}$$

and the $p \times n_{col}$ block matrix

$$\boldsymbol{\Phi}^{(p)} \coloneqq (\boldsymbol{B}_{0,e} \mid \boldsymbol{B}_{1,e-1} \mid \dots \mid \boldsymbol{B}_{e,0})$$

where

$$n_{col} \coloneqq (e+1)r - (p-1) \cdot \frac{q^{e+1}-1}{q-1}$$

For $p = 2$, this is the same matrix Φ considered in [23, section 3]. Note that $\boldsymbol{\Phi}^{(p)}$ is designed such that its rank is 1 if we substitute any column of $\boldsymbol{A}_{\mathbb{F}_{q^m}}$ for the indeterminates $X_i^{(j)}$. For, substituting $X_i^{(j)} \mapsto (YX^i)^{q^j}$, we obtain each row from the previous one by multiplying by X^{q^e}. This is essentially why we obtain polynomials of higher-order vanishing from it, as we shall see in the following.

Lemma 4. *The $p \times p$ minors of $\boldsymbol{\Phi}^{(p)}$ lie in the kernel of $\psi_{m,r}^{(p)}$. They are linearly independent over $\mathbb{F}_{q^m}$.*

Proof. Note that by construction, the matrix $\boldsymbol{\Phi}^{(p)}$ has rank 1 if we substitute $X_i^{(j)} \mapsto (YX^i)^{q^j}$. Thus, for $2 \le \ell \le p$, any $\ell \times \ell$ minor of $\boldsymbol{\Phi}^{(p)}$ vanishes when evaluated at $X_i^{(j)} \mapsto (YX^i)^{q^j}$.

By the rule for differentiating a determinant, any partial derivative of order $p-\ell$ for $2 \le \ell \le p$ of a $p \times p$ minor of $\boldsymbol{\Phi}^{(p)}$ is a linear combination of $\ell \times \ell$ minors of $\boldsymbol{\Phi}^{(p)}$, so it vanishes when evaluated at $X_i^{(j)} \mapsto (YX^i)^{q^j}$. Hence any such $p \times p$ minor is in the kernel of $\psi_{m,r}^{(p)}$.

We prove linear independence by induction on p and first introduce some notation. Given a set $S \subset \{1, \dots, n_{col}\}$ with $|S| = p$, we denote by $\boldsymbol{\Phi}_S^{(p)}$ the $p \times p$ submatrix formed by the columns of $\boldsymbol{\Phi}^{(p)}$ with indices in S.

The entries in any fixed row of $\boldsymbol{\Phi}^{(p)}$ are distinct indeterminates of the polynomial ring and therefore linearly independent. Now suppose that $p \ge 2$ and suppose for contradiction that the $p \times p$ minors are not linearly independent. Consider a linear relation

$$\sum_{i=1}^{\ell} \lambda_i \cdot \det\left(\boldsymbol{\Phi}_{S_i}^{(p)}\right)$$

where $\ell \ge 1$, $\lambda_i \in \mathbb{F}_{q^m} \backslash \{0\}$ for all $1 \le i \le \ell$ and $S_i \subset \{1, \dots, n_{col}\}$ are distinct subsets of size p. Let $a = \max_i \left(\bigcup S_i\right)$ and suppose without loss of generality that

$a \in S_i$ if and only if $\ell \geq i > \ell'$. Note that the indeterminate (let us call it $X_b^{(c)}$) in the bottom rightmost entry of the matrices $\boldsymbol{\Phi}_{S_i}^{(p)}$ for $i > \ell'$ only occurs in this entry and nowhere else in the matrices $\boldsymbol{\Phi}_{S_i}^{(p)}$ for $1 \leq i \leq \ell$. Hence using Laplace expansion of the determinants by the last row and comparing coefficients of $X_b^{(c)}$, we deduce that

$$\sum_{i=\ell'+1}^{\ell} \lambda_i \cdot \det\left(\hat{\boldsymbol{\Phi}}_{S_i \setminus \{a\}}^{(p)}\right) = 0$$

where $\hat{\boldsymbol{\Phi}}^{(p)}$ denotes the matrix $\boldsymbol{\Phi}^{(p)}$ with the last row deleted. By induction, we deduce that $\lambda_i = 0$ for $\ell \geq i > \ell'$, contradicting the assumption that $\lambda_i \neq 0$ for all i. This concludes our proof. □

Proposition 6. *The dimension of the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ is at least $\binom{n_{col}}{p}$.*

Proof. As explained above, we may work over the extension field $\mathbb{F}_{q^m}$ and prove the claim for $\tilde{\varphi}_{\boldsymbol{A}_{\mathbb{F}_{q^m}}}^{(p)}$. By Lemma 4, we know that the $p \times p$ minors of $\boldsymbol{\Phi}^{(p)}$ are linearly independent and are in the kernel of $\varphi_{\boldsymbol{A}_{\mathbb{F}_{q^m}}}^{(p)}$. We must show that under the change of variable that transforms $\boldsymbol{A}_{\mathbb{F}_{q^m}}$ to reduced row echelon form, these minors are mapped to elements in $U_{mr}^{(p)}$, i.e. to polynomials where each indeterminate occurs to degree at most 1.

Let $\boldsymbol{S} \in \mathbb{F}_{q^m}^{mr \times mr}$ be an invertible matrix such that $\boldsymbol{S}\boldsymbol{A}_{\mathbb{F}_{q^m}}$ is in reduced row echelon form. Let $\tilde{\boldsymbol{\Phi}}^{(p)}$ be the matrix obtained from $\boldsymbol{\Phi}^{(p)}$ by applying the map

$$\alpha_{\boldsymbol{S}^{-1}} : \mathbb{F}_{q^m}\left[X_i^{(j)}\right] \to \mathbb{F}_{q^m}\left[X_i^{(j)}\right], \quad f(\boldsymbol{X}) \mapsto f(\boldsymbol{S}^{-1}\boldsymbol{X})$$

to each entry. By definition, if we evaluate each entry of $\tilde{\boldsymbol{\Phi}}^{(p)}$ at a column in $\boldsymbol{S}\boldsymbol{A}_{\mathbb{F}_{q^m}}$, we obtain the same matrix as evaluating each entry in $\boldsymbol{\Phi}^{(p)}$ at the corresponding column in $\boldsymbol{A}_{\mathbb{F}_{q^m}}$.

Since we assumed that $\boldsymbol{A}$ has full rank, $\boldsymbol{S}\boldsymbol{A}_{\mathbb{F}_{q^m}}$ contains the standard basis vectors as columns. Fix a $p \times p$ submatrix $\boldsymbol{M}$ of $\tilde{\boldsymbol{\Phi}}^{(p)}$. Its entries are linear polynomials in the indeterminates. Consider a fixed but arbitrary indeterminant $X_{i_0}^{(j_0)}$. Since $\tilde{\boldsymbol{\Phi}}^{(p)}$ and therefore $\boldsymbol{M}$ has rank at most 1 when evaluated at the standard basis vector $\boldsymbol{e}_{i_0}$, the matrix formed by the coefficients of X_{i_0} in $\boldsymbol{M}$ has rank at most 1. This means that using elementary column operations on $\boldsymbol{M}$, one can transform it such that X_{i_0} occurs only in a single column. The determinant of such a matrix is clearly linear in $X_{i_0}^{(j_0)}$. Since elementary column operations do not alter the determinant, this implies that the same holds for $\boldsymbol{M}$. □

Note that $\boldsymbol{\Phi}^{(p)}$ must have at least p columns for our construction to yield an element in the kernel. Furthermore, the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ is generally larger than the lower bound from Proposition 6. For our purposes, Proposition 6 in combination with the considerations on shortening (see the next section) will be sufficient.

5.2 The effect of shortening

We briefly recall the definition of a shortened code.

Definition 9. *Let $C \subset \mathbb{F}_q^n$ be a linear code and $i \in \{1, \dots, n\}$. The* code shortened in position i *is given by $\{\boldsymbol{c} \in C \mid c_i = 0\}$.*

We may of course puncture such a shortened code in its ith position essentially without losing any information since the ith coordinate of any codeword in the shortened code is 0 anyway. Note that the dimension decreases by at most 1 when shortening a code.

It turns out that shortening alternant and Goppa codes in not too many positions (we make this more precise below) preserves enough of the code structure that the kernel of $\tilde{\varphi}^{(p)}$ is still non-trivial. The advantage of shortening is that it decreases the dimension of our code, which in turn significantly decreases the complexity of computing the dimension of the kernel of $\tilde{\varphi}^{(p)}$. Note that the following result is very similar to [23, Theorem 1].

Proposition 7. *Let $\boldsymbol{A}$ be the parity check matrix of a proper alternant code and let $\boldsymbol{A}_{s_*}$ be the matrix obtained from $\boldsymbol{A}$ by shortening in $s_* := n_{col} - p$ places. Then the kernel of $\tilde{\varphi}^{(p)}_{\boldsymbol{A}_{s_*}}$ is non-trivial.*

Proof. We may work over the extension field $\mathbb{F}_{q^m}$. Let $\boldsymbol{a}_1, \dots, \boldsymbol{a}_{s_*}$ be the column vectors of $\boldsymbol{A}_{\mathbb{F}_{q^m}}$ in the positions where we shorten. Recall that for any i, the matrix obtained by evaluating all entries of $\boldsymbol{\Phi}^{(p)}$ at $\boldsymbol{a}_i$ has rank 1. Thus, using elementary column operations over $\mathbb{F}_{q^m}$, we may transform $\boldsymbol{\Phi}^{(p)}$ to a column-equivalent matrix such that all entries in the first $n_{col} - s_* = p$ columns vanish when evaluated at any $\boldsymbol{a}_i$. Thus, the $p \times p$ minor obtained from the first p columns corresponds to a non-trivial element in the kernel of $\varphi^{(p)}_{\boldsymbol{A}_{s_*}}$. By the same argument as in the proof of Proposition 6, it follows that this yields a non-trivial element in the kernel of $\tilde{\varphi}^{(p)}_{\boldsymbol{A}_{s_*}}$. □

Our computations suggest that if we shorten a random alternant code in more than s_* places, then the kernel becomes trivial with high probability. See Table 2 and Table 3 for some concrete computations. Thus, the expression for s_* seems to be tight in the sense that it is almost surely the *maximum* number of places that we can shorten without obtaining a trivial kernel. Of course, it may still occur that we can shorten in more places. For example, we will see that in the case of Goppa codes, the above expression for s_* can be improved.

5.3 The special case of Goppa codes

For simplicity, we restrict our discussion to the case $q = 2$. We will use [23, Proposition 7], which we reproduce here for convenience:

Table 2. Dimension of the kernel of $\tilde{\varphi}^{(3)}_{\boldsymbol{A}}$ where $\boldsymbol{A}$ is the parity check matrix of a shortened alternant code with $q = 2$, $m = 10$, $r = 7$ for different numbers s of shortened positions. Indeed, we have $s_* = 5$ in this case.

s	0	1	2	3	4	5	6
n	≥ 839	≥ 821	≥ 802	≥ 783	≥ 763	≥ 742	≥ 722
$\dim\left(\ker\left(\tilde{\varphi}^{(3)}\right)\right)$	950	600	350	150	40	10	0

Table 3. Dimension of the kernel of $\tilde{\varphi}^{(3)}_{\boldsymbol{A}}$ where $\boldsymbol{A}$ is the parity check matrix of a shortened alternant code with $q = 5$, $m = 4$, $r = 9$ for different numbers s of shortened positions. Indeed, we have $s_* = 4$ in this case.

s	0	1	2	3	4	5
n	≥ 229	≥ 220	≥ 211	≥ 201	≥ 191	≥ 182
$\dim\left(\ker\left(\tilde{\varphi}^{(3)}\right)\right)$	224	108	40	16	4	0

Proposition 8. *Let $\boldsymbol{x} \in \mathbb{F}_{2^m}^n$ have pairwise distinct entries and $C = Gop(\boldsymbol{x}, g)^\perp$ be the dual of a proper binary Goppa code with square-free Goppa polynomial g of degree t. Set $\boldsymbol{y} = g(\boldsymbol{x})^{-1}$. Then*

$$C_{\mathbb{F}_{q^m}} = \bigoplus_{i=0}^{m/2-1} GRS_{2t}(\boldsymbol{x}^{4^i}, \boldsymbol{y}^{4^i}), \text{ or}$$

$$C_{\mathbb{F}_{q^m}} = \left(\bigoplus_{i=0}^{(m-1)/2-1} GRS_{2t}(\boldsymbol{x}^{4^i}, \boldsymbol{y}^{4^i})\right) \oplus GRS_t(\boldsymbol{x}^{2^{m-1}}, \boldsymbol{y}^{2^{m-1}})$$

depending on whether m is even or odd.

Corollary 3. *Let $\boldsymbol{x} \in \mathbb{F}_{2^m}^n$ have pairwise distinct entries and $C = Gop(\boldsymbol{x}, g)^\perp$ be the dual of a proper binary Goppa code with square-free Goppa polynomial g of degree t. Let $\hat{e} \coloneqq \left\lfloor \log_4\left(\frac{2t-1}{p-1}\right)\right\rfloor$ and $\boldsymbol{A}_{\hat{s}_*}$ be the generator matrix of the code obtained by shortening C in*

$$\hat{s}_* \coloneqq 2(\hat{e}+1)t - (p-1) \cdot \frac{4^{\hat{e}+1}-1}{3} - p$$

places. Then the kernel of $\tilde{\varphi}^{(p)}_{\boldsymbol{A}_{s_*}}$ is non-trivial.

Proof. Same as Proposition 7, with $\boldsymbol{\Phi}^{(p)}$ adapted to be compatible with Proposition 8. ◻

Remark 6. Experiments show that this expression for $\hat{s}_*$ is not tight, i.e. that one can usually shorten more while still maintaining a non-trivial kernel. For example, for the parameter set in Table 4, the formula in Corollary 3 implies we can shorten in 15 positions. As we can see from the results, we can shorten a bit more (namely in 17 positions) and still obtain a non-trivial kernel.

Table 4. Dimension of the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(3)}$ where $\boldsymbol{A}$ is the parity check matrix of a shortened binary Goppa code with $m = 10$, $r = 7$ for different numbers s of shortened positions.

s	13	14	15	16	17	18
n	≥ 574	≥ 561	≥ 546	≥ 529	≥ 512	≥ 496
$\dim\left(\ker\left(\tilde{\varphi}^{(3)}\right)\right)$	560	270	100	40	10	0

6 The distinguisher

Suppose now that for fixed $m, r, n \in \mathbb{N}$, a $\mathbb{F}_q$-code C is randomly sampled from one of

(i) the set of proper alternant (or binary Goppa) codes over $\mathbb{F}_q$ with field extension degree m, degree r and length n
(ii) the set of linear $[n, k]$-codes over $\mathbb{F}_q$ where $k = mr$

and one is to decide whether C comes from (i) or (ii). Our distinguisher proceeds as follows. Initially, we set $p = 2$. Then:

1. Let s be the integer defined in Proposition 7 (if we want to distinguish alternant codes) or the integer defined in Corollary 3 (in the case we want to distinguish binary Goppa codes).
   If $s < 0$, then abort. Otherwise, shorten the code C in s positions and let $\boldsymbol{A} \in \mathbb{F}_q^{k' \times n'}$ be a parity check matrix of the shortened code where $k' = k - s$ and $n' = n - s$.
2. Determine the smallest $\hat{n}$ such that the degree p coefficient of
$$\left[(1+t)^{k'} \cdot \left(1-t^2\right)^{\hat{n}}\right]^+$$
   is zero. If $\hat{n} > n' - k'$, set $p = p + 1$ and return to step 1.
3. Determine whether or not the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ is trivial. One way is to determine
$$\beta^{(p)} := \dim_{\mathbb{F}_q}\left(\ker\left(\tilde{\varphi}_{\boldsymbol{A}}^{(p)}\right)\right)$$
   by computing a matrix representation of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ and applying Gaussian elimination. Another way is to compute a Gröbner basis of the ideal $\mathfrak{L}_{\tilde{\boldsymbol{A}}}$ (defined in Equation 4) with respect to some monomial order and to check whether all monomials of degree p in $\tilde{R}_{k'}$ occur as leading monomials or not.

In step 1, we shorten in just enough positions that we still expect $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ to have a non-trivial kernel in the alternant or Goppa case. Note that in the case of binary Goppa codes, by Remark 6, we can usually shorten a bit more than the amount predicted by Corollary 3. Thus, we are overly strict in the Goppa case by aborting in case the expression for s is negative. In practice, we could try to increase the amount of shortening a bit more in the Goppa case before aborting. Assuming Heuristic 1 holds, step 2 ensures that the kernel of $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ is indeed trivial in the random case with high probability.

Step 3 dominates the complexity of our distinguisher. The matrix representing $\tilde{\varphi}_{\boldsymbol{A}}^{(p)}$ has $\binom{k-s}{p}$ rows. Therefore, it increases rapidly as p grows. Given any concrete parameter set, it is easy to determine whether our distinguisher is able to successfully distinguish or not, and which amount of shortening and which value of p are required. This allows us to determine that our distinguisher does indeed work e.g. for Classic McEliece parameter sets, and to estimate its complexity in this case in subsection 6.2. Giving a closed expression in terms of the parameters n and k for the size of p required seems to be difficult though. For, it is hard to provide a closed formula in terms of n and k that determines the lowest degree where the coefficient in $(1+t)^k \cdot \left(1-t^2\right)^{\tilde{n}}$ is negative. In other words, it is hard to determine a closed formula for the degree of regularity of a semi-regular sequence. For any given concrete parameters, this is easy to compute of course. [4] provides asymptotic formulas and [14] determines bounds in some cases.

6.1 Some example computations

We present some experimental results for our distinguisher for different parameter sets below. We conducted each experiment several times. The values of $\beta^{(p)}$ that we obtained in below experiments in the case of randomly chosen codes almost always matched the prediction of Heuristic 1. There were only very slight deviations. In the random case in Table 5, we sometimes obtained dimension 2 for $n = 952$ instead of 1 as indicated in the table, while the prediction from Heuristic 1 would be 0. Similarly, in Table 6, we sometimes obtained $\beta^{(3)} = 1$ instead of $\beta^{(3)} = 0$ for $n = 3401$ in the random case. We observed no more significant deviations. This provides further evidence for the validity of Heuristic Assumption 1.

Example 2. We consider the same parameters as in [23, Example 2] which seeks to distinguish binary Goppa codes with parameters $m = 10$, $n = 1024$ and $r = 10$ from random codes. For these parameters, our distinguisher works at degree $p = 3$. Corollary 3 predicts that we can shorten the code in 27 positions. Using this, we experimentally find that $\beta^{(3)} = 350$ in the Goppa case, while $\beta^{(3)} = 0$ in the random case. See Table 5 for more detailed experimental results. Our computation required a few GB of RAM and took a couple of minutes of computation.

Example 3. We consider binary Goppa codes with parameters $m = 12$, $r = 17$. This is significantly larger than Example 2. Shortening in 64 positions, we obtain the results displayed in Table 6 for different values of n. Thus, we see that we can distinguish at degree $p = 3$ for $n \geq 3400$. Note that the formula in Corollary 3 predicts that we can shorten in 57 positions for these parameter sets and $p = 3$. This is another example showing that Corollary 3 is not quite tight.

Table 5. Results for distinguishing binary Goppa and random codes with $m = 10$, $r = 10$ at $p = 3$, shortening in 27 positions.

n	≥ 953	952	951	950	949	948	947	946
$\beta^{(3)}$ (Goppa)	350	350	350	350	350	350	365	438
$\beta^{(3)}$ (random)	0	1	73	146	219	292	365	438

Table 6. Results for distinguishing binary Goppa and random codes with $m = 12$, $r = 17$ at $p = 3$, shortening in 64 positions.

n	≥ 3401	3400	3399	3398	3397
$\beta^{(3)}$ (Goppa)	240	240	280	420	560
$\beta^{(3)}$ (random)	0	140	280	420	560

Example 4. We apply the distinguisher at degree $p = 4$ to the parameter set $m = 8$, $r = 6$ to distinguish binary Goppa codes and random codes. The experimental results can be found in Table 7, indicating that we can distinguish for $n \geq 196$.

Table 7. Results for distinguishing binary Goppa and random codes with $m = 8$, $r = 6$ at $p = 4$, shortening in 5 positions.

n	≥ 197	196	195	194
$\beta^{(4)}$ (Goppa)	1120	1120	1400	2157
$\beta^{(4)}$ (random)	0	644	1400	2157

Example 5. We provide an example for distinguishing alternant codes in characteristic 3 by applying the distinguisher at $p = 4$ to the parameter set $m = 5$, $r = 9$. See Table 8 for the experimental results. We remark that the values of $\beta^{(4)}$ we obtain in the random case for various n precisely match the expression in Heuristic 1, confirming our conjecture that it holds in all characteristics. In this regard, also note that the values occurring for $\beta^{(4)}$ in the random case match those from Table 7 in the random case (only shifted in n) because after shortening, we consider polynomials in 43 indeterminates in both cases.

6.2 Performance for Classic McEliece parameters

Consider a binary Goppa code $\mathrm{Gop}(x, g)$ with field extension degree m, length n and Goppa degree $\deg(g) = r$ with irreducible Goppa polynomial g. For the Classic McEliece parameters, Table 9 presents the choice of p and the required shortening in order to distinguish Goppa codes with these parameters from random linear codes. These estimates hold under the assumption that Heuristic 1 is valid. Here N and M denote the number of rows and columns, respectively,

Table 8. Results for distinguishing alternant and random codes with $q = 3$, $m = 5$, $r = 9$ at $p = 4$, shortening in 2 positions.

n	≥ 194	193	192	191
$\beta^{(4)}$ (alternant)	5	644	1400	2157
$\beta^{(4)}$ (random)	0	644	1400	2157

of the matrix representing $\tilde{\varphi}^{(p)}$ whose rank our distinguisher computes. By Remark 6, it is likely we can actually shorten in more positions than indicated in the table, so N and M can probably be made a bit smaller. Naively assuming that computation of rank of a matrix with N rows can be done in complexity N^{ω}, for the smallest Classic McEliece parameter set we estimate a complexity of about 2^{332} which is lower than the complexity reported in [23]. Of course, this is only a rough estimate. Note that in our approach, it is sufficient to determine whether $\tilde{\varphi}^{(p)}_{\boldsymbol{A}}$ has non-trivial kernel or not. So we could also use Wiedemann's algorithm [26] as suggested in [23].

Table 9. Required degree p and number of shortened positions $\hat{s}_*$ for distinguishing Goppa codes with indicated Classic McEliece parameters from random linear codes. N and M denote the number of rows and columns, respectively, of the matrix representing $\tilde{\varphi}^{(p)}$.

(n, r, m)	$(3488, 64, 12)$	$(4608, 96, 13)$	$(6688, 128, 13)$	$(6960, 119, 13)$	$(8192, 128, 13)$
$\hat{s}_*$	123	91	175	217	271
p	23	51	57	44	41
$\log_2(N)$	139.6	297.5	344.7	274.8	262.8
$\log_2(M)$	146.6	305.2	352.7	282.6	270.7

6.3 Relation to the syzygy distinguisher

Let C be a $[n, k]$-code over $\mathbb{F}_q$ with generator matrix $\boldsymbol{B}$ whose columns we denote by $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n$ and rows by $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_k$. [23, Proposition 14] allows one to compare the syzygy distinguisher with the approach presented here. [23, Proposition 14 and Lemma 14] describes the syzygy module as $\ker \partial|_T$ where, in our notation,

$$\partial : \left(\bigwedge\nolimits^{p-1} R_k^{(1)}\right) \otimes C \longrightarrow \left(\bigwedge\nolimits^{p-2} R_k^{(1)}\right) \otimes \mathbb{F}_q^n$$

and T is the subspace spanned by basis vectors

$$\boldsymbol{e}_{i_1,\ldots,i_{p-1};l} := (X_{i_1} \wedge \cdots \wedge X_{i_{p-1}}) \otimes \boldsymbol{c}_l$$

with $1 \leq i_1 < \cdots < i_{p-1} \leq k$ and $1 \leq l \leq i_{p-1}$.

$\partial|_T$ is given explicitly by the images of the basis vectors

$$\boldsymbol{e}_{i_1,\dots,i_{p-1};l} \mapsto \sum_{j=1}^{p-1}(-1)^j X_{i_1}\wedge\cdots\hat{X_{i_j}}\cdots\wedge X_{i_{p-1}}\otimes[\mathrm{ev}_{\boldsymbol{b}_1}(X_{i_j}X_l),\dots,\mathrm{ev}_{\boldsymbol{b}_n}(X_{i_j}X_l)]$$

where notation $\hat{X_{i_j}}$ means that X_{i_j} is omitted and $\mathrm{ev}_{\boldsymbol{c}}$ denotes evaluation of a polynomial at a vector $\boldsymbol{c}$. We now assume that $\boldsymbol{B}$ is in reduced row-echelon form and that $\boldsymbol{b}_s \in \mathbb{F}_q^k$ is equal to the s-th standard basis vector for $s \le k$. Let $T' \subset T$ be spanned by all basis vectors $\boldsymbol{e}_{i_1,\dots,i_{p-1};l}$ in T such that l is different from all $i_1,\dots,i_{p-1}$. Then one can see that $\ker\partial|_T$ is equal to the kernel of the map

$$\partial''\colon T' \to \left(\bigwedge^{p-2} R_k^{(1)}\right)\otimes\mathbb{F}_q^{n-k}$$

defined by

$$\begin{aligned}\boldsymbol{e}_{i_1,\dots,i_{p-1};l} \mapsto \sum_{j=1}^{p-1}((-1)^j X_{i_1}\wedge\cdots\hat{X_{i_j}}\cdots\wedge X_{i_{p-1}}&\\ \otimes[\mathrm{ev}_{\boldsymbol{b}_{k+1}}(X_{i_j}X_l),\dots,\mathrm{ev}_{\boldsymbol{b}_n}(X_{i_j}X_l)])&\end{aligned}$$

If we modify $\tilde{\varphi}_{\boldsymbol{B}}^{(p)}$ (as defined in Definition 6) to consider derivatives of order equal to $p-2$ only, this can be understood as the map defined on monomials μ as

$$\begin{aligned}U_k^{(p)} &\to \left(\bigwedge^{p-2} R_k^{(1)}\right)\otimes\mathbb{F}_q^{n-k},\\ \mu &\mapsto \sum_{\substack{\nu \text{ with } \nu|\mu,\\ \deg(\nu)=p-2}} \nu\otimes\left[\mathrm{ev}_{\boldsymbol{b}_{k+1}}\left(\frac{\mu}{\nu}\right),\dots,\mathrm{ev}_{\boldsymbol{b}_n}\left(\frac{\mu}{\nu}\right)\right]\end{aligned}$$

This map is very similar to the map ∂'' by considering $\nu = X_{i_1}\wedge\cdots\hat{X_{i_j}}\cdots\wedge X_{i_{p-1}}$ and $\mu = \nu X_l X_{i_j}$. However, there are differences. For each μ the image is a sum over $\binom{p}{2}$ terms, whereas ∂'' is a sum over $p-1$ terms. The number of basis elements in ∂'' of the form $\boldsymbol{e}_{i_1,\dots,i_{p-1};l}$ is equal to $(p-1)\binom{k}{p}$. In the end, it does not seem likely that the distinguisher presented here can be reduced to the distinguisher from [23]. But both distinguishers consider similar elements in the space $\left(\bigwedge^{p-2}R_k^{(1)}\right)\otimes\mathbb{F}_q^{n-k}$.

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